Design Patterns Analysis - Flight Reservation System

- Shubham Goel (2023101076)
- Swam Singla (2023101134)
- Saksham Chitkara (2023101021)
- Shreyas Mehta (2023101059)
- Srinivas Chitta (2025900001)

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Observer Pattern

Overview

The Observer Pattern defines a one-to-many dependency between objects so that when one object changes state, all its dependents are notified automatically.

Current State Analysis

Challenge: The system lacks notification mechanisms for important events:

- Flight booking confirmation
- Capacity changes
- Payment processing status
- Schedule updates

Application and Reasoning

Why Apply This Pattern?

- Multiple entities need to react to booking events (Customer, ScheduledFlight, PaymentProcessor)
- Decouples event producers from consumers
- Enables flexible, real-time notifications
- Supports logging, auditing, and analytics without modifying core classes

Benefits:

- ✅ Loose coupling between components
- ✅ Dynamic subscription/unsubscription at runtime
- ✅ Supports multiple observers

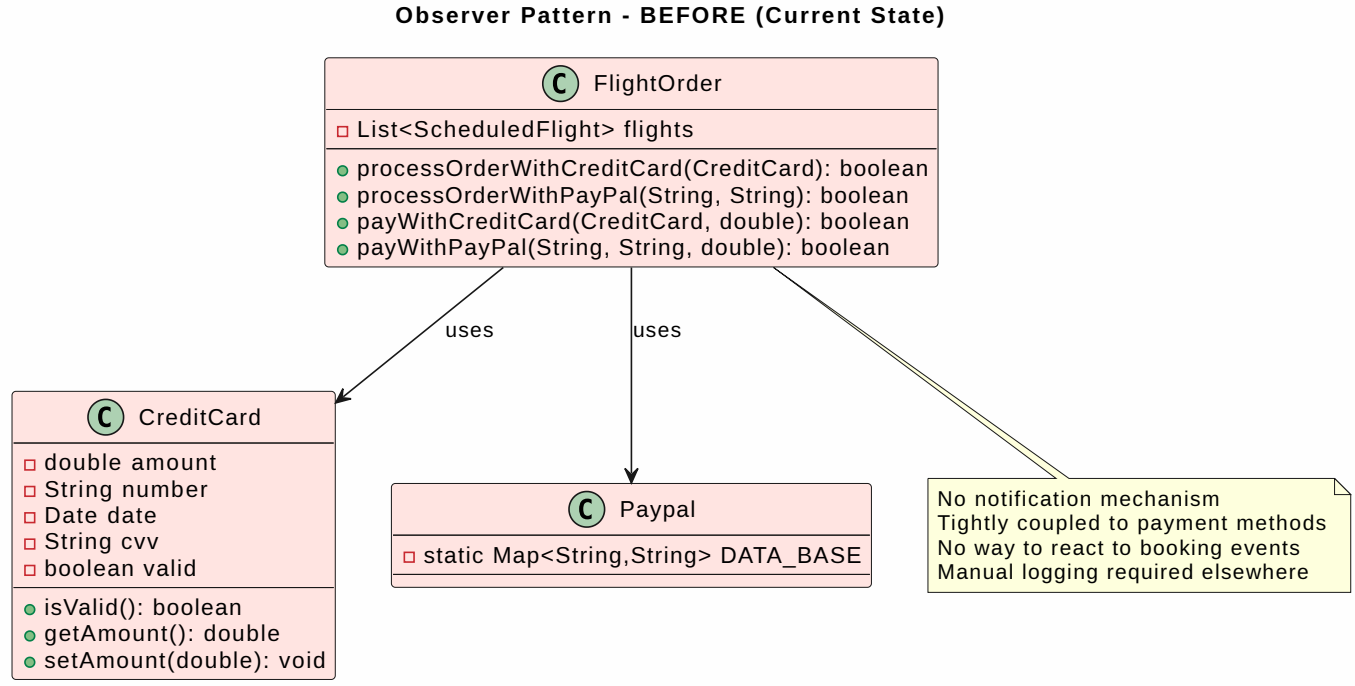
- ✅ Enables email/SMS notifications, logging, audit trails
- ✅ Easy to extend with new notification types

Drawbacks:

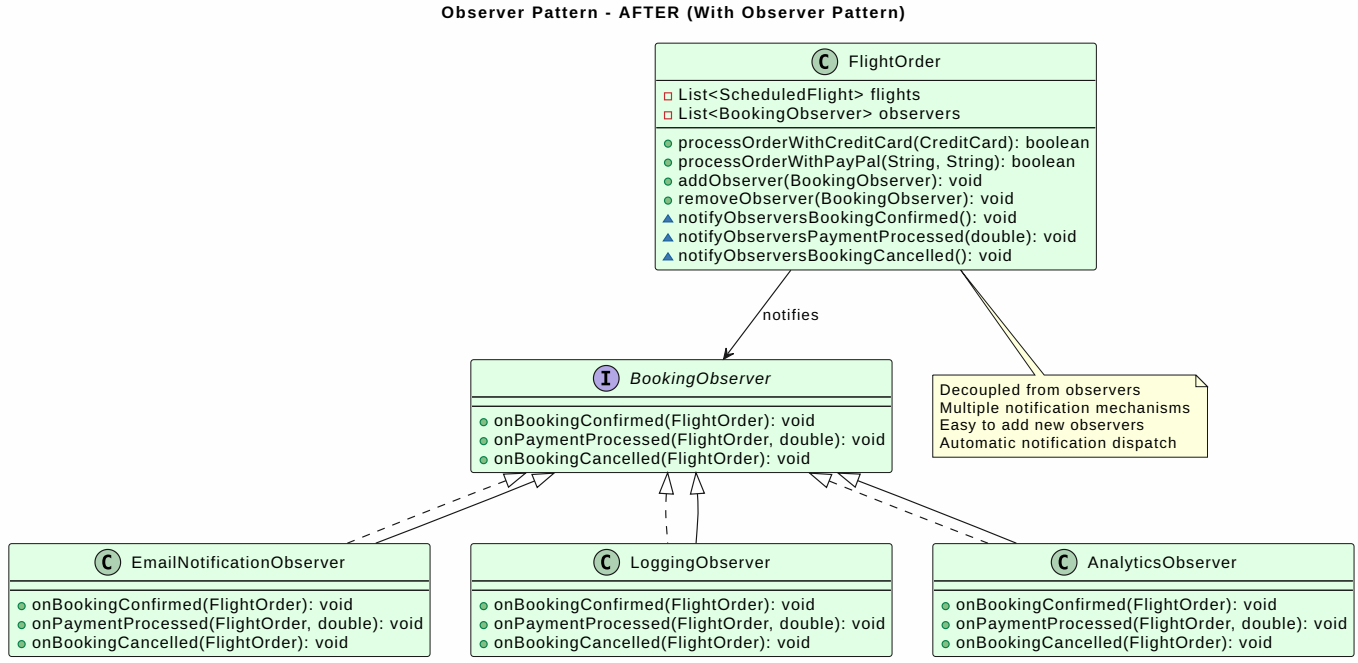
- ❌ Performance overhead with many observers
- ❌ Notification order is unpredictable
- ❌ Memory overhead for storing observer references
- ❌ Potential for memory leaks if observers not unsubscribed

Class Diagram

Before Pattern:



After Pattern:



Implementation Example

```java
// Observer Interface
public interface BookingObserver {
    void onBookingConfirmed(FlightOrder order);
    void onPaymentProcessed(FlightOrder order, double amount);
    void onBookingCancelled(FlightOrder order);
}

// Concrete Observer - Email Notification
public class EmailNotificationObserver implements BookingObserver {
    @Override
    public void onBookingConfirmed(FlightOrder order) {
        String email = order.getCustomer().getEmail();
        System.out.println("Sending confirmation email to: " + email);
        // Send email logic
    }

    @Override
    public void onPaymentProcessed(FlightOrder order, double amount) {
        String email = order.getCustomer().getEmail();
        System.out.println("Sending payment receipt to: " + email +
                           " for amount: $" + amount);
    }

    @Override
    public void onBookingCancelled(FlightOrder order) {
        System.out.println("Sending cancellation notice to: " +
                           order.getCustomer().getEmail());
    }
}

// Concrete Observer - Logging
public class LoggingObserver implements BookingObserver {
    @Override
    public void onBookingConfirmed(FlightOrder order) {
        System.out.println("[LOG] Booking confirmed: " + order.getId());
    }

    @Override
    public void onPaymentProcessed(FlightOrder order, double amount) {
        System.out.println("[LOG] Payment processed: $" + amount +
                           " for order " + order.getId());
    }

    @Override
    public void onBookingCancelled(FlightOrder order) {
        System.out.println("[LOG] Booking cancelled: " + order.getId());
    }
}

// Subject - Modified FlightOrder
public class FlightOrder extends Order {
    private final List<ScheduledFlight> flights;
```

```java
    private List<BookingObserver> observers = new ArrayList<>();

    // ... existing code ...

    public void addObserver(BookingObserver observer) {
        observers.add(observer);
    }

    public void removeObserver(BookingObserver observer) {
        observers.remove(observer);
    }

    private void notifyObserversBookingConfirmed() {
        observers.forEach(observer -> observer.onBookingConfirmed(this));
    }

    private void notifyObserversPaymentProcessed(double amount) {
        observers.forEach(observer -> observer.onPaymentProcessed(this,
amount));
    }

    private void notifyObserversBookingCancelled() {
        observers.forEach(observer -> observer.onBookingCancelled(this));
    }

    public boolean processOrderWithCreditCard(CreditCard creditCard)
            throws IllegalStateException {
        if (isClosed()) {
            return true;
        }
        if (!cardIsPresentAndValid(creditCard)) {
            throw new IllegalStateException("Payment information is
invalid.");
        }
        boolean isPaid = payWithCreditCard(creditCard, this.getPrice());
        if (isPaid) {
            this.setClosed();
            notifyObserversPaymentProcessed(this.getPrice());
            notifyObserversBookingConfirmed();
        }
        return isPaid;
    }
}

// Usage
public class FlightOrderClient {
    public static void main(String[] args) {
        FlightOrder order = new FlightOrder(flights);

        // Subscribe observers
        order.addObserver(new EmailNotificationObserver());
        order.addObserver(new LoggingObserver());

        // Process payment - will automatically notify all observers
```

```
            order.processOrderWithCreditCard(creditCard);
        }
    }
```

Factory Pattern

Overview

The Factory Pattern provides an interface for creating objects without specifying their exact classes, promoting loose coupling and flexibility.

Current State Analysis

Challenge: Aircraft creation uses switch statements with type checking in multiple places:

```
// Current approach scattered across codebase
if (aircraft instanceof PassengerPlane) {
    // handle PassengerPlane
} else if (aircraft instanceof Helicopter) {
    // handle Helicopter
}
```

Application and Reasoning

Why Apply This Pattern?

- Centralized aircraft creation logic
- Easy to add new aircraft types (PassengerPlane, Helicopter, PassengerDrone)
- Reduces duplication of type checking logic
- Simplifies capacity and crew management
- Enables polymorphic handling of different aircraft

Benefits:

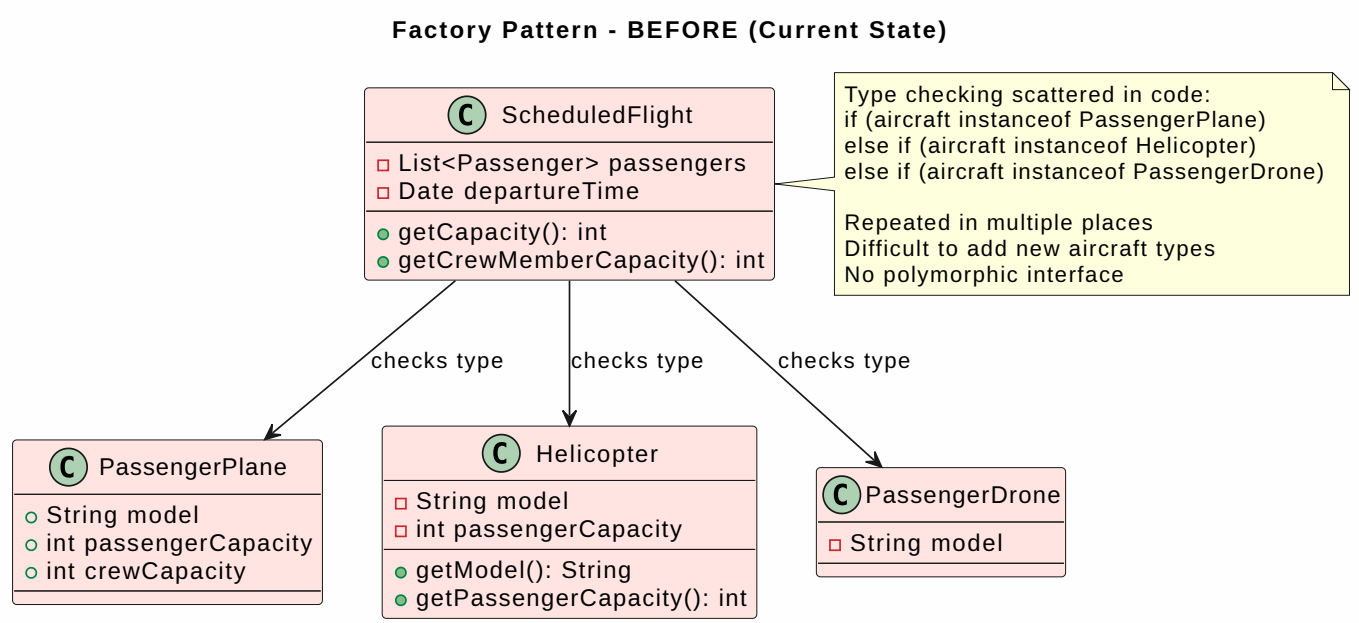
- ✅ Encapsulates object creation logic
- ✅ Eliminates switch/if-else statements for type checking
- ✅ Easy to add new aircraft types
- ✅ Single responsibility for aircraft creation
- ✅ Improves code maintainability
- ✅ Supports polymorphism

Drawbacks:

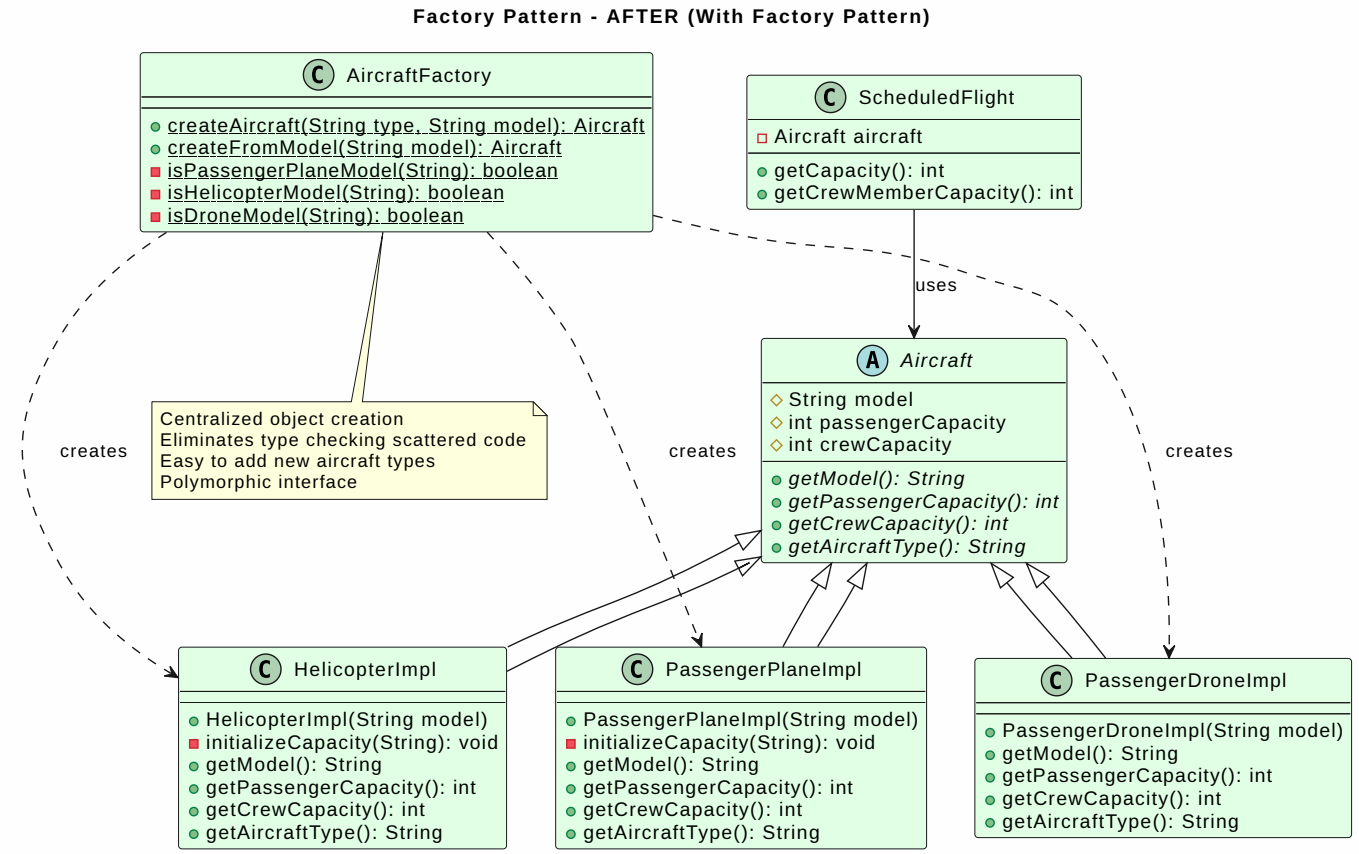
- ❌ Adds more classes to the codebase
- ❌ Small overhead for simple object creation
- ❌ May be overkill for simple systems
- ❌ Requires refactoring existing code

Class Diagram

Before Pattern:



After Pattern:



Implementation Example

```
// Aircraft Base Interface/Class
public abstract class Aircraft {
    protected String model;
    protected int passengerCapacity;
    protected int crewCapacity;
```

```
    public abstract String getModel();
    public abstract int getPassengerCapacity();
    public abstract int getCrewCapacity();
    public abstract String getAircraftType();
}

// Concrete Aircraft Classes (refactored)
public class PassengerPlaneImpl extends Aircraft {
    public PassengerPlaneImpl(String model) {
        this.model = model;
        initializeCapacity(model);
    }

    private void initializeCapacity(String model) {
        switch (model) {
            case "A380":
                this.passengerCapacity = 500;
                this.crewCapacity = 42;
                break;
            case "A350":
                this.passengerCapacity = 320;
                this.crewCapacity = 40;
                break;
            case "Embraer 190":
                this.passengerCapacity = 25;
                this.crewCapacity = 5;
                break;
            case "Antonov AN2":
                this.passengerCapacity = 15;
                this.crewCapacity = 3;
                break;
            default:
                throw new IllegalArgumentException("Model not recognized: "
+ model);
        }
    }

    @Override
    public String getModel() { return model; }

    @Override
    public int getPassengerCapacity() { return passengerCapacity; }

    @Override
    public int getCrewCapacity() { return crewCapacity; }

    @Override
    public String getAircraftType() { return "PassengerPlane"; }
}

public class HelicopterImpl extends Aircraft {
    public HelicopterImpl(String model) {
        this.model = model;
```

```
        initializeCapacity(model);
    }

    private void initializeCapacity(String model) {
        switch (model) {
            case "H1":
                this.passengerCapacity = 4;
                this.crewCapacity = 1;
                break;
            case "H2":
                this.passengerCapacity = 6;
                this.crewCapacity = 1;
                break;
            default:
                throw new IllegalArgumentException("Model not recognized: "
+ model);
        }
    }

    @Override
    public String getModel() { return model; }

    @Override
    public int getPassengerCapacity() { return passengerCapacity; }

    @Override
    public int getCrewCapacity() { return crewCapacity; }

    @Override
    public String getAircraftType() { return "Helicopter"; }
}

public class PassengerDroneImpl extends Aircraft {
    public PassengerDroneImpl(String model) {
        if (model.equals("HypaHype")) {
            this.model = model;
            this.passengerCapacity = 4;
            this.crewCapacity = 0;
        } else {
            throw new IllegalArgumentException("Model not recognized: " +
model);
        }
    }

    @Override
    public String getModel() { return model; }

    @Override
    public int getPassengerCapacity() { return passengerCapacity; }

    @Override
    public int getCrewCapacity() { return crewCapacity; }

    @Override
```

```java
    public String getAircraftType() { return "PassengerDrone"; }
}

// Aircraft Factory
public class AircraftFactory {

    public static Aircraft createAircraft(String type, String model)
            throws IllegalArgumentException {
        switch (type.toLowerCase()) {
            case "passengerplane":
                return new PassengerPlaneImpl(model);
            case "helicopter":
                return new HelicopterImpl(model);
            case "passengerdrone":
                return new PassengerDroneImpl(model);
            default:
                throw new IllegalArgumentException("Unknown aircraft type:
" + type);
        }
    }

    // Convenience method for direct model-based creation
    public static Aircraft createFromModel(String model) {
        if (isPassengerPlaneModel(model)) {
            return new PassengerPlaneImpl(model);
        } else if (isHelicopterModel(model)) {
            return new HelicopterImpl(model);
        } else if (isDroneModel(model)) {
            return new PassengerDroneImpl(model);
        } else {
            throw new IllegalArgumentException("Unknown model: " + model);
        }
    }

    private static boolean isPassengerPlaneModel(String model) {
        return model.matches("A380|A350|Embraer 190|Antonov AN2");
    }

    private static boolean isHelicopterModel(String model) {
        return model.matches("H1|H2");
    }

    private static boolean isDroneModel(String model) {
        return model.equals("HypaHype");
    }
}

// Usage in ScheduledFlight (refactored)
public class ScheduledFlight extends Flight {
    // ... existing code ...

    public int getCapacity() throws NoSuchFieldException {
        if (this.aircraft instanceof Aircraft) {
            return ((Aircraft) this.aircraft).getPassengerCapacity();
```

```
        }
        throw new NoSuchFieldException("Aircraft information unavailable");
    }

    public int getCrewMemberCapacity() throws NoSuchFieldException {
        if (this.aircraft instanceof Aircraft) {
            return ((Aircraft) this.aircraft).getCrewCapacity();
        }
        throw new NoSuchFieldException("Aircraft information unavailable");
    }
}

// Client Usage
public class RunnerWithFactory {
    public static void main(String[] args) {
        // Create aircraft using factory
        Aircraft plane = AircraftFactory.createAircraft("passengerplane",
"A380");
        Aircraft helicopter = AircraftFactory.createAircraft("helicopter",
"H1");
        Aircraft drone = AircraftFactory.createFromModel("HypaHype");

        // Create flights with factory-created aircraft
        Flight flight1 = new Flight(1, berlinAirport, frankfurtAirport,
plane);
        Flight flight2 = new Flight(2, jfkAirport, berlinAirport,
helicopter);
    }
}
```

Adapter Pattern

Overview

The Adapter Pattern converts the interface of a class into another interface that clients expect, allowing incompatible interfaces to work together (also called Wrapper).

Current State Analysis

Challenge: Payment methods (CreditCard and PayPal) have different interfaces:

```
// CreditCard has: isValid(), getAmount(), setAmount()
// PayPal has: static DATA_BASE map with different structure

// Usage is inconsistent and scattered in FlightOrder
boolean isPaid = payWithCreditCard(creditCard, amount);  // direct method
boolean isPaid = payWithPayPal(email, password, amount); // different
signature
```

Application and Reasoning

Why Apply This Pattern?

- Unifies different payment method interfaces
- Allows new payment methods to be added without modifying core order logic
- Encapsulates payment processing details
- Enables polymorphic treatment of all payment methods
- Reduces code duplication in order processing

Benefits:

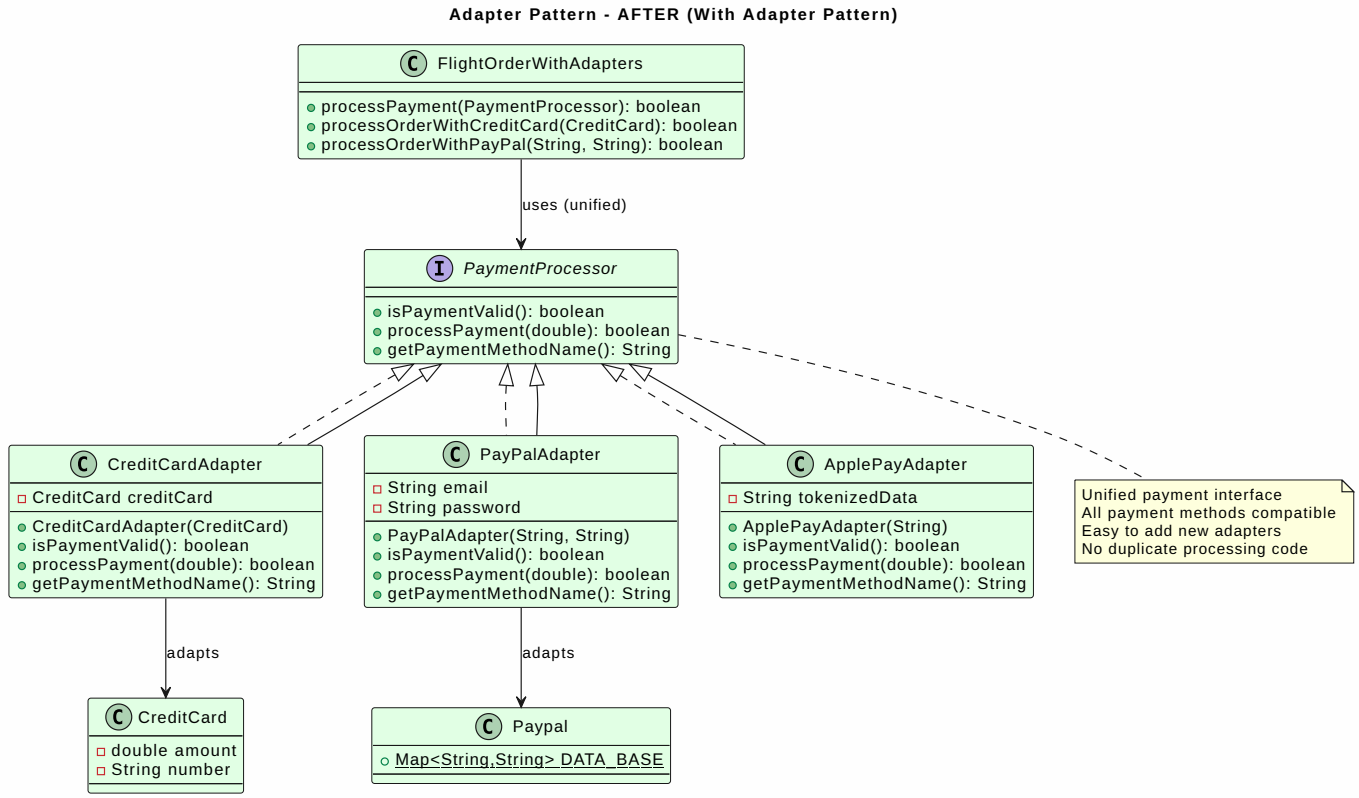
- ✅ Makes incompatible interfaces compatible
- ✅ Reduces code duplication
- ✅ Single unified payment interface
- ✅ Easy to add new payment methods
- ✅ Decouples payment logic from order logic
- ✅ Better testability and flexibility

Drawbacks:

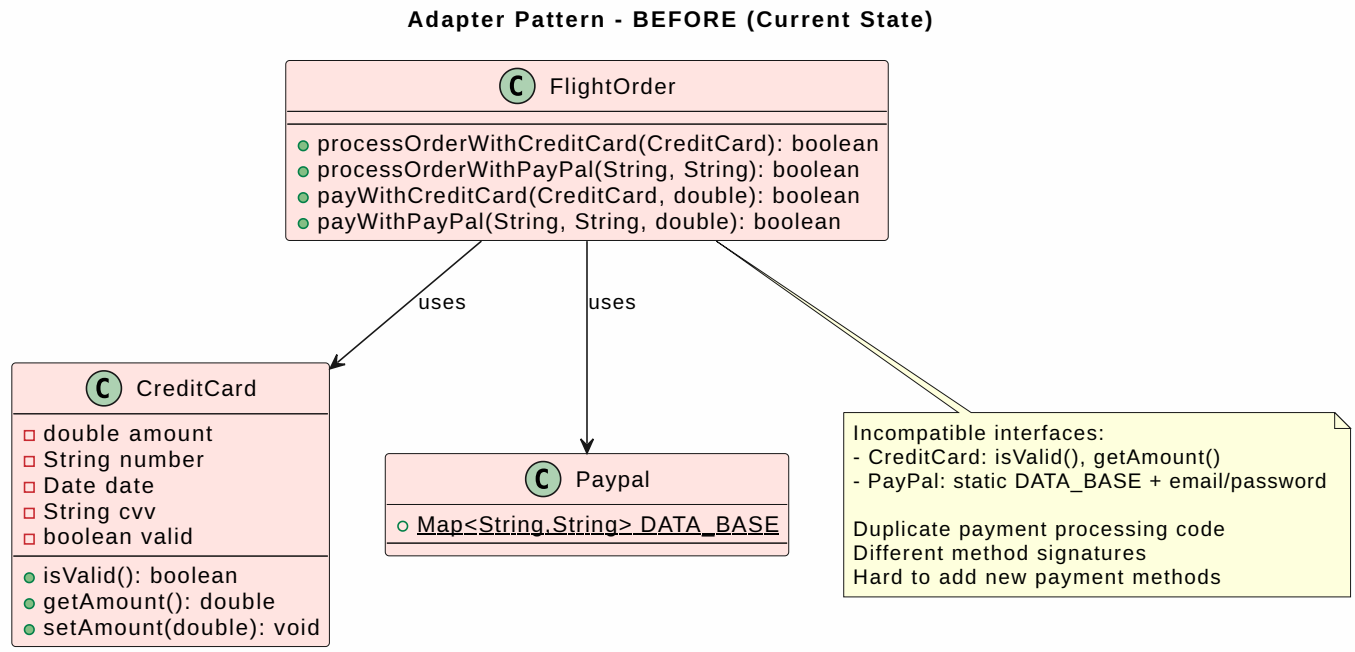
- ❌ Adds additional layer of abstraction
- ❌ Slight performance overhead
- ❌ May overcomplicate simple scenarios
- ❌ Requires understanding of adapter hierarchy

Class Diagram

Before Pattern:



After Pattern:



Implementation Example

```
// Common Payment Interface (Target Interface)
public interface PaymentProcessor {
    boolean isPaymentValid();
    boolean processPayment(double amount) throws IllegalStateException;
    String getPaymentMethodName();
}

// Adapter for Credit Card
public class CreditCardAdapter implements PaymentProcessor {
    private final CreditCard creditCard;

    public CreditCardAdapter(CreditCard creditCard) {
        this.creditCard = creditCard;
    }

    @Override
    public boolean isPaymentValid() {
        return creditCard != null && creditCard.isValid();
    }

    @Override
    public boolean processPayment(double amount) throws
IllegalStateException {
        if (!isPaymentValid()) {
            throw new IllegalStateException("Invalid credit card");
        }

        double remainingAmount = creditCard.getAmount() - amount;
        if (remainingAmount < 0) {
            System.out.printf("Card limit reached - Balance: %f%n",
remainingAmount);
```

```java
            throw new IllegalStateException("Insufficient card balance");
        }

        System.out.println("Processing payment of $" + amount +
                           " via Credit Card");
        creditCard.setAmount(remainingAmount);
        return true;
    }

    @Override
    public String getPaymentMethodName() {
        return "Credit Card";
    }
}

// Adapter for PayPal
public class PayPalAdapter implements PaymentProcessor {
    private final String email;
    private final String password;

    public PayPalAdapter(String email, String password) {
        this.email = email;
        this.password = password;
    }

    @Override
    public boolean isPaymentValid() {
        return email != null &&
               password != null &&
               email.equals(Paypal.DATA_BASE.get(password));
    }

    @Override
    public boolean processPayment(double amount) throws
IllegalStateException {
        if (!isPaymentValid()) {
            throw new IllegalStateException("Invalid PayPal credentials");
        }

        System.out.println("Processing payment of $" + amount +
                           " via PayPal");
        // PayPal processing logic
        return true;
    }

    @Override
    public String getPaymentMethodName() {
        return "PayPal";
    }
}

// Adapter for Apple Pay/Future payment methods
public class ApplePayAdapter implements PaymentProcessor {
    private final String tokenizedData;
```

```java
    public ApplePayAdapter(String tokenizedData) {
        this.tokenizedData = tokenizedData;
    }

    @Override
    public boolean isPaymentValid() {
        return tokenizedData != null && tokenizedData.length() > 0;
    }

    @Override
    public boolean processPayment(double amount) throws
IllegalStateException {
        if (!isPaymentValid()) {
            throw new IllegalStateException("Invalid Apple Pay token");
        }

        System.out.println("Processing payment of $" + amount +
                           " via Apple Pay");
        return true;
    }

    @Override
    public String getPaymentMethodName() {
        return "Apple Pay";
    }
}

// Refactored FlightOrder using Adapters
public class FlightOrderWithAdapters extends Order {
    private final List<ScheduledFlight> flights;
    static List<String> noFlyList = Arrays.asList("Peter", "Johannes");

    public FlightOrderWithAdapters(List<ScheduledFlight> flights) {
        this.flights = flights;
    }

    public List<ScheduledFlight> getScheduledFlights() {
        return flights;
    }

    // Unified payment processing using adapters
    public boolean processPayment(PaymentProcessor processor)
            throws IllegalStateException {
        if (isClosed()) {
            return true; // Already processed
        }

        if (!processor.isPaymentValid()) {
            throw new IllegalStateException(
                "Payment validation failed for " +
                processor.getPaymentMethodName());
        }
```

```
        boolean isPaid = processor.processPayment(this.getPrice());
        if (isPaid) {
            this.setClosed();
            System.out.println("Order " + this.getId() +
                               " successfully paid using " +
                               processor.getPaymentMethodName());
        }
        return isPaid;
    }

    // Legacy methods - now use adapters internally
    public boolean processOrderWithCreditCard(CreditCard creditCard)
            throws IllegalStateException {
        PaymentProcessor processor = new CreditCardAdapter(creditCard);
        return processPayment(processor);
    }

    public boolean processOrderWithPayPal(String email, String password)
            throws IllegalStateException {
        PaymentProcessor processor = new PayPalAdapter(email, password);
        return processPayment(processor);
    }
}

// Client Usage
public class PaymentProcessingClient {
    public static void main(String[] args) {
        FlightOrderWithAdapters order = new
FlightOrderWithAdapters(flights);

        // Process with different payment methods using same interface
        CreditCard card = new CreditCard("4532123456789",
                                         new Date(), "123");
        PaymentProcessor cardProcessor = new CreditCardAdapter(card);
        order.processPayment(cardProcessor);

        // Add new payment method without modifying FlightOrder
        PaymentProcessor applePayProcessor =
            new ApplePayAdapter("tokenizedApplePayData");
        order.processPayment(applePayProcessor);
    }
}
```

Builder Pattern

Overview

The Builder Pattern constructs complex objects step by step, separating the construction process from the object's representation, resulting in more readable and flexible code.

Current State Analysis

Challenge: Creating a FlightOrder requires multiple sequential steps with error-prone parameter passing:

```
// Current approach - difficult to manage
FlightOrder order = new FlightOrder(flights);
order.setCustomer(customer);
order.setPrice(price);
order.setPassengers(passengers);
// Price must be calculated separately
// Sequence is important but not enforced
```

Application and Reasoning

Why Apply This Pattern?

- Complex multi-step order creation process
- Many optional and required parameters
- Improves code readability and maintainability
- Validates object state before creation
- Allows different construction strategies
- Makes object immutable after creation

Benefits:

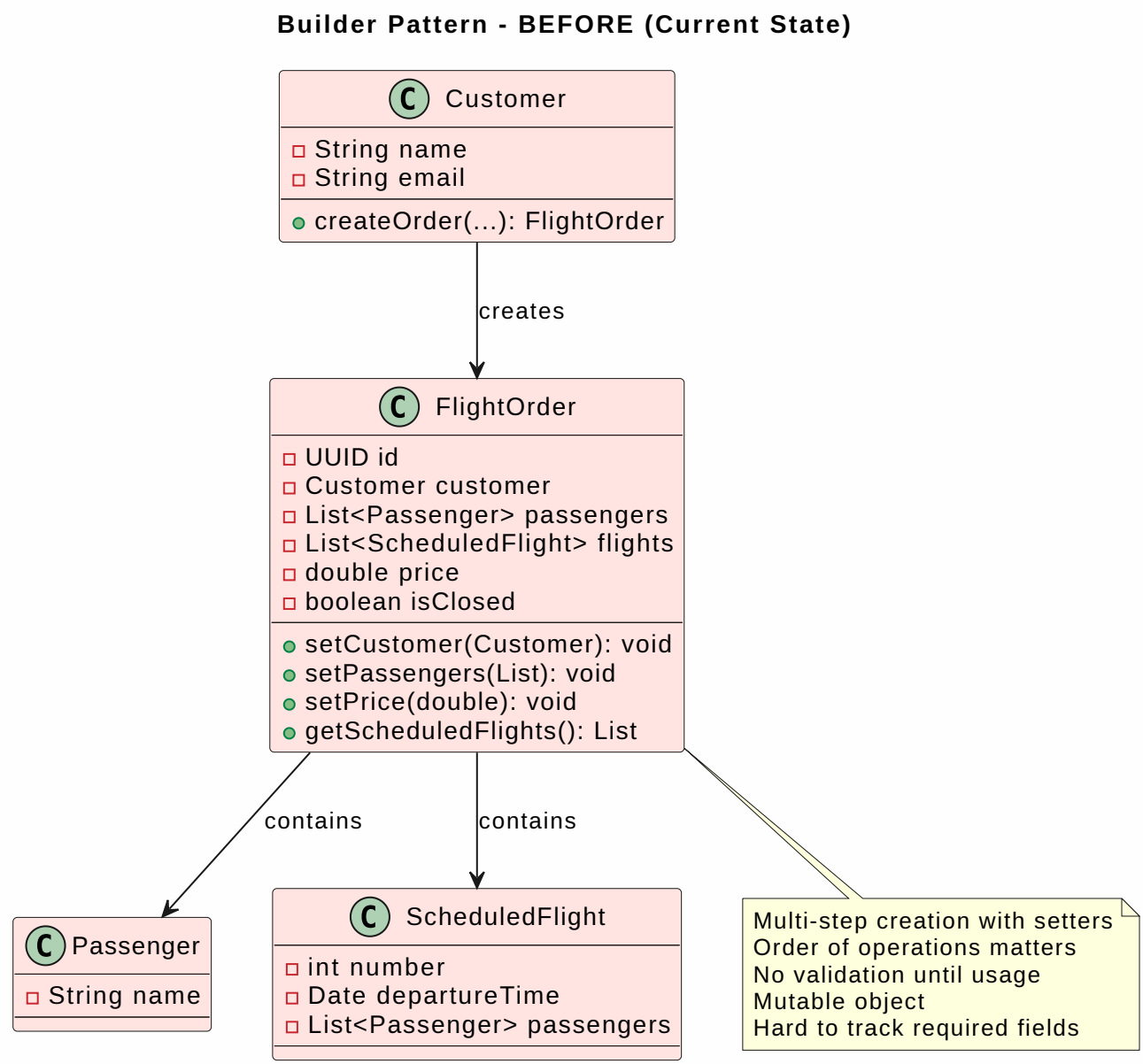
- ✅ Makes complex object construction more readable
- ✅ Separates construction logic from object representation
- ✅ Supports optional parameters elegantly
- ✅ Enforces object consistency before creation
- ✅ Provides fluent interface
- ✅ Easier to understand parameter values

Drawbacks:

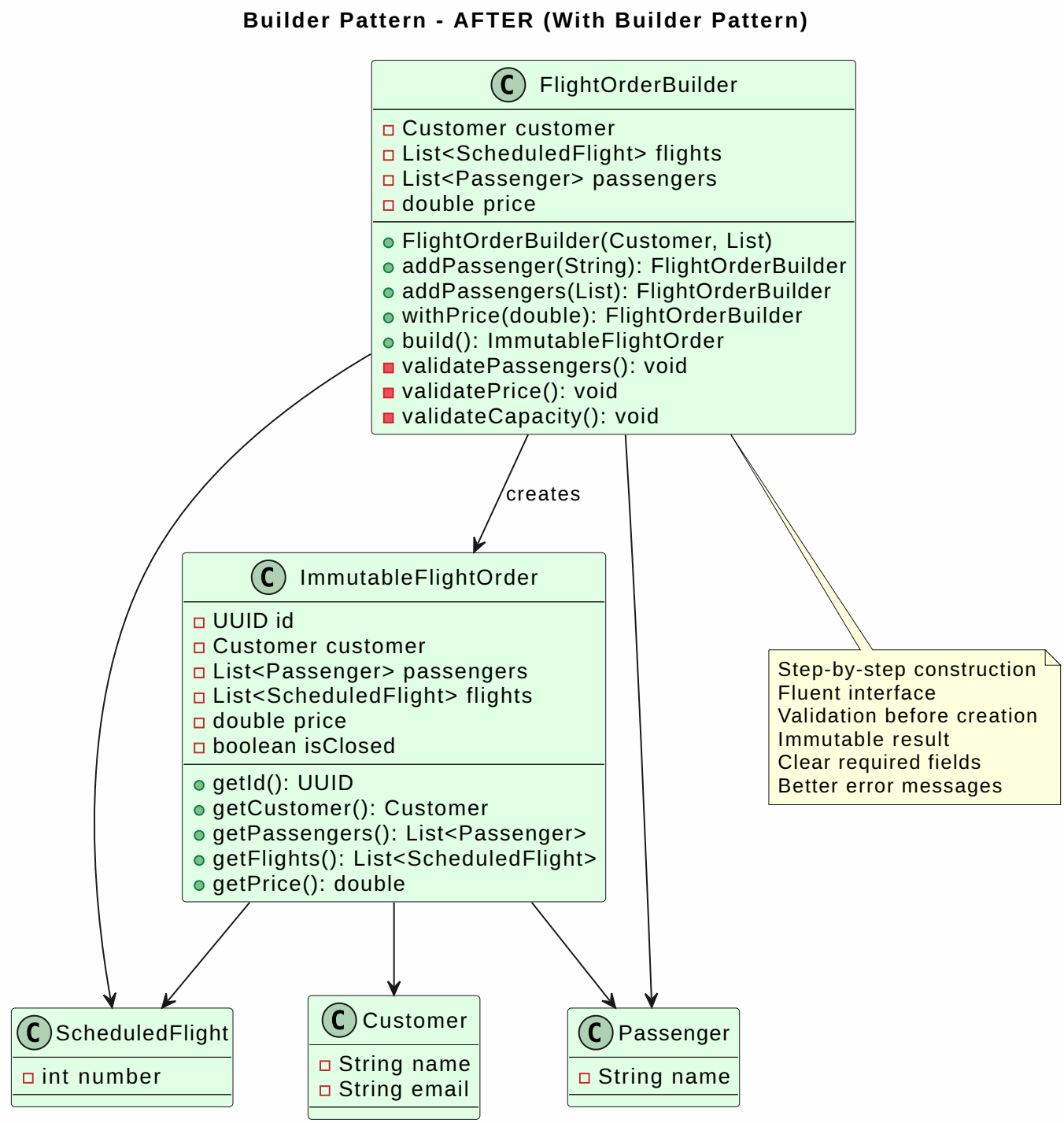
- ❌ Creates additional builder classes
- ❌ Verbose for simple objects
- ❌ Extra memory overhead
- ❌ More code to maintain

Class Diagram

Before Pattern:


Builder Pattern - BEFORE (Current State)
Customer
String name
String email
createOrder(...): FlightOrder
creates
FlightOrder
UUID id
Customer customer
List<Passenger> passengers
List<ScheduledFlight> flights
double price
boolean isClosed
setCustomer(Customer): void
setPassengers(List): void
setPrice(double): void
getScheduledFlights(): List
contains
contains
Passenger
String name
ScheduledFlight
int number
Date departureTime
List<Passenger> passengers
Multi-step creation with setters
Order of operations matters
No validation until usage
Mutable object
Hard to track required fields

After Pattern:



Implementation Example

```
// Immutable FlightOrder after creation
public class ImmutableFlightOrder {
    private final UUID id;
    private final Customer customer;
    private final List<Passenger> passengers;
    private final List<ScheduledFlight> flights;
    private final double price;
    private final boolean isClosed;

    private ImmutableFlightOrder(FlightOrderBuilder builder) {
```

```java
        this.id = UUID.randomUUID();
        this.customer = builder.customer;
        this.passengers = new ArrayList<>(builder.passengers);
        this.flights = new ArrayList<>(builder.flights);
        this.price = builder.price;
        this.isClosed = false;
    }

    // Static builder class
    public static class FlightOrderBuilder {
        private final Customer customer;
        private final List<ScheduledFlight> flights;
        private final List<Passenger> passengers;
        private double price = 0.0;

        public FlightOrderBuilder(Customer customer,
                                  List<ScheduledFlight> flights) {
            if (customer == null || flights == null || flights.isEmpty()) {
                throw new IllegalArgumentException(
                    "Customer and flights are required");
            }
            this.customer = customer;
            this.flights = flights;
            this.passengers = new ArrayList<>();
        }

        // Add passenger by name
        public FlightOrderBuilder addPassenger(String passengerName) {
            if (passengerName == null || passengerName.isEmpty()) {
                throw new IllegalArgumentException(
                    "Passenger name cannot be empty");
            }
            this.passengers.add(new Passenger(passengerName));
            return this;
        }

        // Add multiple passengers
        public FlightOrderBuilder addPassengers(List<String> names) {
            if (names == null || names.isEmpty()) {
                throw new IllegalArgumentException(
                    "At least one passenger is required");
            }
            names.forEach(this::addPassenger);
            return this;
        }

        // Set price
        public FlightOrderBuilder withPrice(double price) {
            if (price <= 0) {
                throw new IllegalArgumentException(
                    "Price must be greater than 0");
            }
            this.price = price;
            return this;
```

```
        }

        // Validate and build
        public ImmutableFlightOrder build() {
            if (passengers.isEmpty()) {
                throw new IllegalStateException(
                    "At least one passenger is required");
            }

            if (price == 0.0) {
                throw new IllegalStateException(
                    "Price must be set before building order");
            }

            // Validate customers on no-fly list
            if (FlightOrder.getNoFlyList().contains(customer.getName())) {
                throw new IllegalStateException(
                    "Customer is on the no-fly list");
            }

            // Validate passengers on no-fly list
            String noFlyPassenger = passengers.stream()
                .map(Passenger::getName)
                .filter(name -> FlightOrder.getNoFlyList().contains(name))
                .findFirst()
                .orElse(null);

            if (noFlyPassenger != null) {
                throw new IllegalStateException(
                    "Passenger " + noFlyPassenger +
                    " is on the no-fly list");
            }

            // Validate capacity
            try {
                for (ScheduledFlight flight : flights) {
                    if (flight.getAvailableCapacity() < passengers.size())
{
                        throw new IllegalStateException(
                            "Insufficient capacity on flight " +
                            flight.getNumber());
                    }
                }
            } catch (NoSuchFieldException e) {
                throw new IllegalStateException(
                    "Cannot determine flight capacity");
            }

            return new ImmutableFlightOrder(this);
        }
    }

    // Getters
    public UUID getId() { return id; }
```

```
    public Customer getCustomer() { return customer; }
    public List<Passenger> getPassengers() { return
List.copyOf(passengers); }
    public List<ScheduledFlight> getFlights() { return
List.copyOf(flights); }
    public double getPrice() { return price; }
    public boolean isClosed() { return isClosed; }
}

// Usage
public class FlightOrderBuilderClient {
    public static void main(String[] args) throws NoSuchFieldException {
        Customer customer = new Customer("Max Mustermann",
"max@example.com");
        List<ScheduledFlight> flights = new ArrayList<>();
        ScheduledFlight flight = new ScheduledFlight(1, berlin, frankfurt,
                                                     plane, new Date());
        flights.add(flight);

        try {
            // Build order step by step
            ImmutableFlightOrder order =
                new ImmutableFlightOrder.FlightOrderBuilder(customer,
flights)
                    .addPassenger("Alice Smith")
                    .addPassenger("Bob Johnson")
                    .withPrice(500.0)
                    .build();

            System.out.println("Order created successfully: " +
order.getId());
            System.out.println("Passengers: " +
order.getPassengers().size());
            System.out.println("Total price: $" + order.getPrice());

        } catch (IllegalStateException e) {
            System.err.println("Failed to build order: " + e.getMessage());
        }
    }
}
```

Strategy Pattern

Overview

The Strategy Pattern defines a family of algorithms, encapsulates each one, and makes them interchangeable. It lets algorithms vary independently from the clients that use them.

Current State Analysis

Challenge: Payment processing logic is tightly coupled within FlightOrder:

```
// Different algorithms mixed in same method
public boolean processOrderWithCreditCard(CreditCard creditCard) { ... }
public boolean processOrderWithPayPal(String email, String password) { ...
}
// Validation and processing logic is duplicated
```

Application and Reasoning

Why Apply This Pattern?

- Different payment processing algorithms (Credit Card, PayPal)
- Runtime selection of payment strategy
- Algorithm-specific validation different for each method
- Easy to add new payment strategies (Apple Pay, Bitcoin, etc.)
- Reduces code duplication and complexity

Benefits:

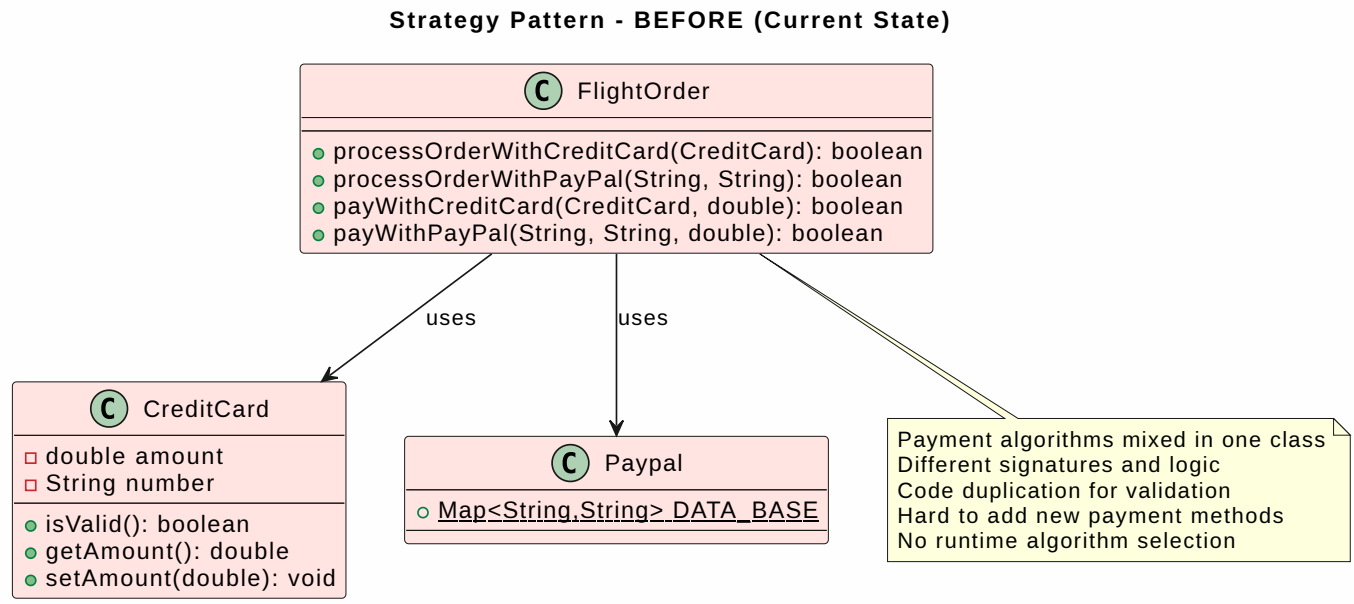
- ✅ Isolates algorithm implementation details
- ✅ Easy to add new payment strategies
- ✅ Runtime selection of strategies
- ✅ Reduces conditional statements
- ✅ Improves testability
- ✅ Single responsibility per strategy
- ✅ Follows Open/Closed Principle

Drawbacks:

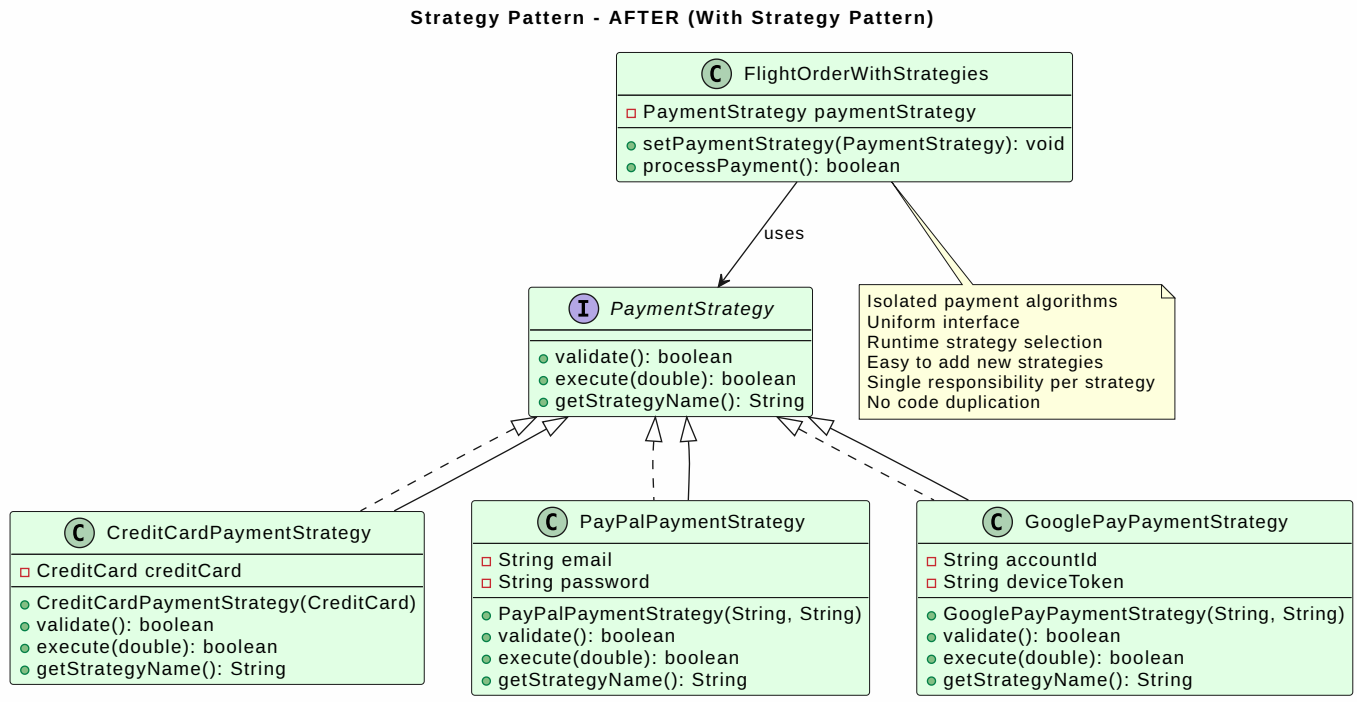
- ❌ More classes and complexity for simple scenarios
- ❌ Strategy selection logic needed elsewhere
- ❌ Overhead of creating strategy objects
- ❌ Potential over-engineering

Class Diagram

Before Pattern:



After Pattern:



Implementation Example

```
// Strategy Interface
public interface PaymentStrategy {
    /**
     * Validates if the payment can be processed
     */
    boolean validate();

    /**
     * Executes the payment processing
     */
    boolean execute(double amount) throws IllegalStateException;
```

```java
    /**
     * Returns the name of the payment strategy
     */
    String getStrategyName();
}

// Concrete Strategy - Credit Card Payment
public class CreditCardPaymentStrategy implements PaymentStrategy {
    private final CreditCard creditCard;

    public CreditCardPaymentStrategy(CreditCard creditCard) {
        this.creditCard = creditCard;
    }

    @Override
    public boolean validate() {
        return creditCard != null && creditCard.isValid();
    }

    @Override
    public boolean execute(double amount) throws IllegalStateException {
        if (!validate()) {
            throw new IllegalStateException("Invalid credit card");
        }

        // Strategy-specific logic
        double remainingAmount = creditCard.getAmount() - amount;
        if (remainingAmount < 0) {
            throw new IllegalStateException(
                "Insufficient card balance. Required: $" + amount +
                ", Available: $" + creditCard.getAmount());
        }

        System.out.println("Charging credit card for $" + amount);
        System.out.println("Remaining balance: $" + remainingAmount);
        creditCard.setAmount(remainingAmount);
        return true;
    }

    @Override
    public String getStrategyName() {
        return "Credit Card Payment";
    }
}

// Concrete Strategy - PayPal Payment
public class PayPalPaymentStrategy implements PaymentStrategy {
    private final String email;
    private final String password;

    public PayPalPaymentStrategy(String email, String password) {
        this.email = email;
        this.password = password;
    }
```

```java
    @Override
    public boolean validate() {
        return email != null &&
               password != null &&
               !email.isEmpty() &&
               email.equals(Paypal.DATA_BASE.get(password));
    }

    @Override
    public boolean execute(double amount) throws IllegalStateException {
        if (!validate()) {
            throw new IllegalStateException(
                "Invalid PayPal credentials");
        }

        // PayPal-specific logic
        System.out.println("Sending payment of $" + amount +
                           " to PayPal...");
        System.out.println("PayPal account: " + email);
        // In real implementation, would call PayPal API
        return true;
    }

    @Override
    public String getStrategyName() {
        return "PayPal Payment";
    }
}

// Concrete Strategy - Google Pay
public class GooglePayPaymentStrategy implements PaymentStrategy {
    private final String accountId;
    private final String deviceToken;

    public GooglePayPaymentStrategy(String accountId, String deviceToken) {
        this.accountId = accountId;
        this.deviceToken = deviceToken;
    }

    @Override
    public boolean validate() {
        return accountId != null &&
               deviceToken != null &&
               !accountId.isEmpty() &&
               !deviceToken.isEmpty();
    }

    @Override
    public boolean execute(double amount) throws IllegalStateException {
        if (!validate()) {
            throw new IllegalStateException(
                "Invalid Google Pay credentials");
        }
```

```java
        System.out.println("Processing Google Pay payment of $" + amount);
        System.out.println("Account ID: " + accountId);
        return true;
    }

    @Override
    public String getStrategyName() {
        return "Google Pay";
    }
}

// Context - FlightOrder using strategies
public class FlightOrderWithStrategies extends Order {
    private final List<ScheduledFlight> flights;
    private PaymentStrategy paymentStrategy;

    public FlightOrderWithStrategies(List<ScheduledFlight> flights) {
        this.flights = flights;
    }

    // Set payment strategy
    public void setPaymentStrategy(PaymentStrategy strategy) {
        this.paymentStrategy = strategy;
        System.out.println("Payment strategy set to: " +
                           strategy.getStrategyName());
    }

    // Process payment using current strategy
    public boolean processPayment() throws IllegalStateException {
        if (isClosed()) {
            return true;
        }

        if (paymentStrategy == null) {
            throw new IllegalStateException(
                "No payment strategy defined");
        }

        System.out.println("Processing order " + this.getId() +
                           " with: " + paymentStrategy.getStrategyName());
        boolean isPaid = paymentStrategy.execute(this.getPrice());

        if (isPaid) {
            this.setClosed();
            System.out.println("Order successfully paid!");
        }

        return isPaid;
    }

    public List<ScheduledFlight> getScheduledFlights() {
        return flights;
    }
```

```
}

// Usage Example
public class PaymentStrategyClient {
    public static void main(String[] args) {
        FlightOrderWithStrategies order =
            new FlightOrderWithStrategies(flights);
        order.setPrice(500.0);

        try {
            // Use Credit Card Strategy
            CreditCard card = new CreditCard("4532123456789010",
                                             new Date(), "123");
            PaymentStrategy cardStrategy =
                new CreditCardPaymentStrategy(card);
            order.setPaymentStrategy(cardStrategy);
            order.processPayment();

            // OR use PayPal Strategy
            // PaymentStrategy paypalStrategy =
            //     new PayPalPaymentStrategy("john@amazon.eu", "qwerty");
            // order.setPaymentStrategy(paypalStrategy);
            // order.processPayment();

            // OR use Google Pay Strategy
            // PaymentStrategy googlePayStrategy =
            //     new GooglePayPaymentStrategy("account123", "device456");
            // order.setPaymentStrategy(googlePayStrategy);
            // order.processPayment();

        } catch (IllegalStateException e) {
            System.err.println("Payment failed: " + e.getMessage());
        }
    }
}
```

Chain of Responsibility Pattern

Overview

The Chain of Responsibility Pattern passes a request along a chain of handlers where each handler decides either to process the request or pass it to the next handler.

Current State Analysis

Challenge: Order validation logic is scattered and mixed:

```
// Validation logic spread across Customer and FlightOrder
private boolean isOrderValid(List<String> passengerNames,
                             List<ScheduledFlight> flights) {
```

```
    boolean valid = true;
    valid = valid && !noFlyList.contains(customer.getName());
    valid = valid && passengerNames.stream()
        .noneMatch(passenger -> noFlyList.contains(passenger));
    valid = valid && flights.stream().allMatch(scheduledFlight -> {
        // Check capacity
    });
    return valid;
}
```

Application and Reasoning

Why Apply This Pattern?

- Multiple validation handlers (no-fly list, capacity, customer history)
- Each validation is independent
- Easy to add new validators without modifying existing code
- Clear separation of validation concerns
- Enables flexible validation chains
- Improves readability

Benefits:

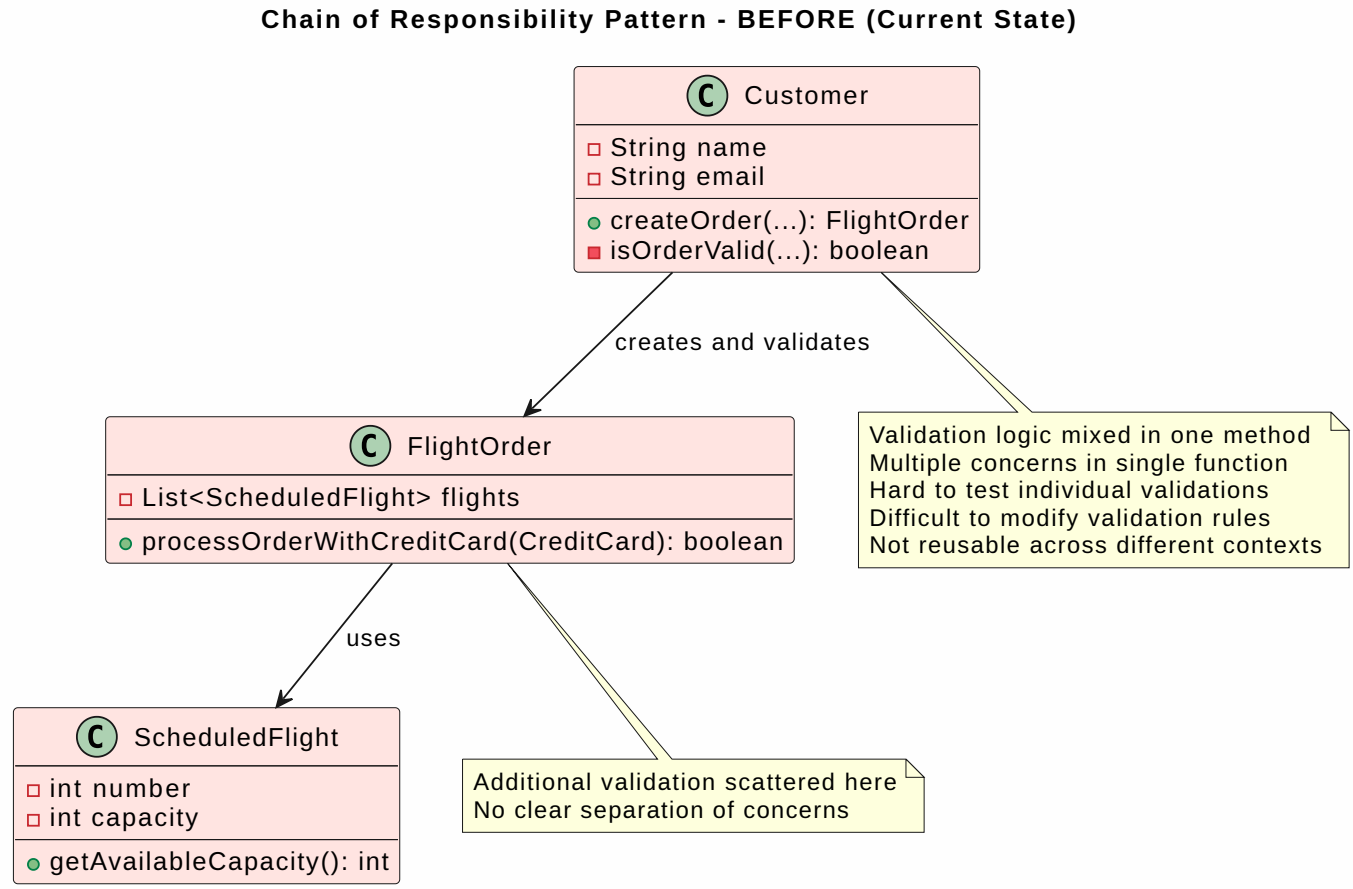
- ✅ Decouples validators from the order
- ✅ Easy to add/remove validators
- ✅ Each validator has single responsibility
- ✅ Validation order can be changed dynamically
- ✅ Improves code readability
- ✅ Follows Open/Closed Principle

Drawbacks:

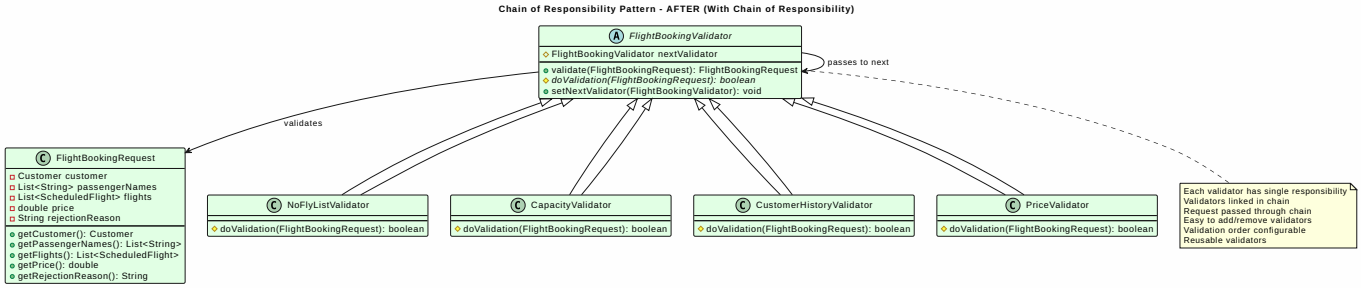
- ❌ Adds more classes and complexity
- ❌ Harder to debug than single validation method
- ❌ Request might not be handled if chain incomplete
- ❌ Performance overhead with multiple handler checks

Class Diagram

Before Pattern:



After Pattern:



Implementation Example

```
// Request object
public class FlightBookingRequest {
    private final Customer customer;
    private final List<String> passengerNames;
    private final List<ScheduledFlight> flights;
    private final double price;
    private String rejectionReason;

    public FlightBookingRequest(Customer customer,
                                List<String> passengerNames,
                                List<ScheduledFlight> flights,
                                double price) {
        this.customer = customer;
        this.passengerNames = passengerNames;
        this.flights = flights;
```

```java
        this.price = price;
    }

    // Getters
    public Customer getCustomer() { return customer; }
    public List<String> getPassengerNames() { return passengerNames; }
    public List<ScheduledFlight> getFlights() { return flights; }
    public double getPrice() { return price; }
    public String getRejectionReason() { return rejectionReason; }
    public void setRejectionReason(String reason) {
        this.rejectionReason = reason;
    }
}

// Handler Interface
public abstract class FlightBookingValidator {
    protected FlightBookingValidator nextValidator;

    public final FlightBookingRequest validate(
            FlightBookingRequest request) {
        if (doValidation(request)) {
            // Validation passed, move to next validator
            if (nextValidator != null) {
                return nextValidator.validate(request);
            }
            // All validators passed
            System.out.println("✓ All validations passed!");
            return request;
        } else {
            // Validation failed
            System.out.println("✗ Validation failed: " +
                               request.getRejectionReason());
            return request;
        }
    }

    // Template method - subclasses implement specific validation
    protected abstract boolean doValidation(FlightBookingRequest request);

    public void setNextValidator(FlightBookingValidator next) {
        this.nextValidator = next;
    }
}

// Concrete Handler 1 - Check No-Fly List
public class NoFlyListValidator extends FlightBookingValidator {

    @Override
    protected boolean doValidation(FlightBookingRequest request) {
        System.out.println("Checking no-fly list...");

        // Check if customer is on no-fly list
        if (FlightOrder.getNoFlyList().contains(
                request.getCustomer().getName())) {
```

```
            request.setRejectionReason(
                "Customer is on the no-fly list");
            return false;
        }

        // Check if any passenger is on no-fly list
        for (String passengerName : request.getPassengerNames()) {
            if (FlightOrder.getNoFlyList().contains(passengerName)) {
                request.setRejectionReason(
                    "Passenger '" + passengerName +
                    "' is on the no-fly list");
                return false;
            }
        }

        System.out.println("✓ No-fly list check passed");
        return true;
    }
}

// Concrete Handler 2 - Check Capacity
public class CapacityValidator extends FlightBookingValidator {

    @Override
    protected boolean doValidation(FlightBookingRequest request) {
        System.out.println("Checking flight capacity...");

        try {
            for (ScheduledFlight flight : request.getFlights()) {
                int requiredCapacity = request.getPassengerNames().size();
                int availableCapacity = flight.getAvailableCapacity();

                if (availableCapacity < requiredCapacity) {
                    request.setRejectionReason(
                        "Insufficient capacity on flight " +
                        flight.getNumber() +
                        ". Required: " + requiredCapacity +
                        ", Available: " + availableCapacity);
                    return false;
                }
            }
        } catch (NoSuchFieldException e) {
            request.setRejectionReason(
                "Cannot determine flight capacity: " + e.getMessage());
            return false;
        }

        System.out.println("✓ Capacity check passed");
        return true;
    }
}

// Concrete Handler 3 - Check Customer History
public class CustomerHistoryValidator extends FlightBookingValidator {
```

```java
    @Override
    protected boolean doValidation(FlightBookingRequest request) {
        System.out.println("Checking customer history...");

        Customer customer = request.getCustomer();

        // Check if customer has unpaid orders
        List<Order> unpaidOrders = customer.getOrders().stream()
            .filter(order -> !order.isClosed())
            .toList();

        if (!unpaidOrders.isEmpty()) {
            request.setRejectionReason(
                "Customer has " + unpaidOrders.size() +
                " unpaid order(s)");
            return false;
        }

        System.out.println("✓ Customer history check passed");
        return true;
    }
}

// Concrete Handler 4 - Check Price Validation
public class PriceValidator extends FlightBookingValidator {

    @Override
    protected boolean doValidation(FlightBookingRequest request) {
        System.out.println("Validating price...");

        if (request.getPrice() <= 0) {
            request.setRejectionReason(
                "Invalid price: must be greater than 0");
            return false;
        }

        // Check if price is reasonable compared to flight distance
        // (simplified example)
        if (request.getPrice() > 100000) {
            request.setRejectionReason(
                "Price appears unreasonably high");
            return false;
        }

        System.out.println("✓ Price validation passed");
        return true;
    }
}

// Usage - Building the chain
public class ChainOfResponsibilityClient {
    public static void main(String[] args) {
        // Build validation chain
```

```
        FlightBookingValidator noFlyValidator =
            new NoFlyListValidator();
        FlightBookingValidator capacityValidator =
            new CapacityValidator();
        FlightBookingValidator historyValidator =
            new CustomerHistoryValidator();
        FlightBookingValidator priceValidator =
            new PriceValidator();

        // Link validators in chain
        noFlyValidator.setNextValidator(capacityValidator);
        capacityValidator.setNextValidator(historyValidator);
        historyValidator.setNextValidator(priceValidator);

        // Create booking request
        Customer customer = new Customer("Max Mustermann",
                                         "max@example.com");
        FlightBookingRequest request = new FlightBookingRequest(
            customer,
            Arrays.asList("Alice Smith", "Bob Johnson"),
            Arrays.asList(flight),
            500.0
        );

        // Execute validation chain
        FlightBookingRequest result = noFlyValidator.validate(request);

        if (result.getRejectionReason() == null) {
            System.out.println("\n✓ Booking approved!");
            // Create the order
        } else {
            System.out.println("\n✗ Booking rejected: " +
                               result.getRejectionReason());
        }
    }
}
```

Summary Comparison Table

Pattern	Best For	Complexity	Scalability	Ease of Use
Observer	Event notifications, reactive updates	Medium	High	Medium
Factory	Object creation, type abstraction	Low	Medium	High
Adapter	Interface incompatibility, legacy code	Low	Medium	High
Builder	Complex object construction	Medium	Medium	Low

Pattern	Best For	Complexity	Scalability	Ease of Use
Strategy	Algorithm selection, payment methods	Medium	High	Medium
Chain of Responsibility	Sequential validation, request handling	Medium	High	Medium

Implementation Recommendations

High Priority (Immediate Implementation)

1. **Strategy Pattern for Payments** - Already partially present; formalize it
2. **Adapter Pattern for Payment Methods** - Unify payment interfaces
3. **Factory Pattern for Aircraft** - Eliminate type checking duplication

Medium Priority (Planned Enhancement)

4. **Chain of Responsibility for Validation** - Cleaner validation logic
5. **Observer Pattern for Notifications** - Add event capabilities

Low Priority (Future Enhancement)

6. **Builder Pattern for Orders** - Complex object construction

Refactoring Strategy

- Start with **Factory Pattern** (lowest risk, immediate benefits)
- Follow with **Strategy Pattern** (payment processing improvement)
- Then **Adapter Pattern** (unified payment interface)
- Implement **Chain of Responsibility** (cleaner validation)
- Add **Observer Pattern** (notification system)
- Finalize with **Builder Pattern** (complex object creation)

Conclusion

The Flight Reservation System is well-suited for implementing multiple design patterns that will:

- Improve code maintainability and readability
- Reduce duplication and complexity
- Enable easier future extensions
- Separate concerns and responsibilities
- Make the codebase more testable and flexible

Each pattern addresses specific pain points in the current implementation while maintaining backward compatibility during gradual adoption.